一、1. A

$P(AB)=P(\overline{A}\overline{B})=1-P(A\cup B)=1-[P(A)+P(B)-P(AB)]$

$\therefore P(B)=1-P(A)=0.4$

2. D.

$2Y-3X\sim N(2\times3-3\times2,\ 4\times4+9\times1)$ 正态分布的可加性

3. C

$Cov(X,Z)=Cov(X,X-Y)=Cov(X,X)-Cov(X,Y)=D(X)-Cov(X,Y)$

$=4-0.5\times2\times2=2.$

4. A

$P\{|X+Y|\geq6\}=P\{|(X+Y)-E(X+Y)|\geq6\}\leq\frac{1}{6^2}D(X+Y)=\frac{1}{12}$

$D(X+Y)=D(X)+D(Y)+2Cov(X,Y)=1+4+2\times(-0.5)\times1\times2=3$

5. A

$\frac{(n-1)S^2}{\sigma^2}=\frac{(10-1)S^2}{9}\sim\chi^2(10-1)\quad\therefore D(S^2)=2\times9=18$

6. A.

σ^2未知时，t检验法

二、1. $\frac{3}{10}$.

抽签模型，无论第几次抽到黑球的概率都是$\frac{3}{10}$，与次序无关。

2. $\frac{3}{5}$

$\rho_{UV}=\frac{Cov(U,V)}{\sqrt{DU}\sqrt{DV}}=\frac{Cov(2X+Y,2X-Y)}{\sqrt{D(2X+Y)}\sqrt{D(2X-Y)}}=\frac{4Cov(X,X)-Cov(Y,Y)}{4D(X)+D(Y)}$ 由方差的性质，方差与协方差

$=\frac{4\lambda-\lambda}{4\lambda+\lambda}=\frac{3}{5}$ 泊松分布的期望、方差.

3. 18.

$\frac{1}{n}\sum_{i=1}^{n}X_i^2\xrightarrow{P}E(X_i^2)=E(X^2)=D(X)+[E(X)]^2=9+3^2$.（辛钦定理的推广）

4. $\left(\bar{X}\pm\frac{S}{\sqrt{2020}}t_{0.05}(2019)\right)$ σ^2未知时的单个正态总体的样本容量和α.

5. D.1

X \ Y	1	2	
1	0.2	0.2	0.4
2	0.2		
3			

$P\{X=1,Y=1\}=P\{X=1\}\cdot P\{Y=1\}=0.4\ P\{X=1\}=0.2$

$P\{X=2\}=0.5.\ \therefore P\{X=2\}=0.5.$ ~~$P\{Y=3|X=0\}$~~

$P\{X=1,Y=3\}=P\{X=1\}\cdot P\{Y=3\}=0.5\cdot P\{Y=3\}=0.1\Rightarrow P\{Y=3\}=0.2$

$\because P\{X=2,Y=3\}=P\{X=2\}\cdot P\{Y=3\}=0.5\times0.2=0.1.$

6. $\frac{1}{4}$

$f(x)=Ae^{-\frac{(x+3)^2}{2\times4}\cdot e^{\frac{y^2}{2}}}$ 利用X与Y独立的条件和边缘分布.

利用X是正态分布.

三、1. 解：设 B：患病，A：呈阳性，则 $P(A|B)=0.99$，$P(A|\bar{B})=0.1$，$P(B)=0.05$

(1) $P(\bar{A})=1-P(A)$ 由全概率公式 $P(A)=P(B)\cdot P(A|B)+P(\bar{B})\cdot P(A|\bar{B})$

$=0.05\times 0.99+0.95\times 0.1=0.1445$

$\therefore P(\bar{A})=0.8555$

(2) $P(B|A)=\dfrac{P(AB)}{P(A)}=\dfrac{P(B)P(A|B)}{P(A)}=\dfrac{0.05\times 0.99}{0.1445}=\dfrac{99}{289}$

2. 设 Y：$X>3$ 的次数，$P\{X>3\}=\int_3^{+\infty}\frac{1}{2}e^{-\frac{x}{2}}dx=e^{-\frac{3}{2}}$.

$\therefore Y\sim b(100,e^{-\frac{3}{2}})$ $P\{Y=10\}=C_{100}^{10}(e^{-\frac{3}{2}})^{10}(1-e^{-\frac{3}{2}})^{100-10}=C_{100}^{10}e^{-15}(1-e^{-\frac{3}{2}})^{90}$

3. (1) $1=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}f(x,y)dxdy=\int_0^{+\infty}\int_0^{+\infty}ke^{-3x-2y}dxdy=\frac{k}{6}$. $\therefore k=6$.

(2) $f_X(x)=\int_{-\infty}^{+\infty}f(x,y)dy=\begin{cases}\int_0^{+\infty}6e^{-3x-2y}dy, & x>0\\ 0, & x\le 0\end{cases}=\begin{cases}3e^{-3x}, & x>0\\ 0, & x\le 0\end{cases}$

$f_Y(y)=\int_{-\infty}^{+\infty}f(x,y)dx=\begin{cases}2e^{-2y}, & y>0\\ 0, & y\le 0\end{cases}$

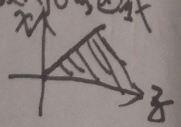
(3) $\because f(x,y)=f_X(x)f_Y(y)$ $\therefore X$ 与 Y 独立.

(4) $f_{X|Y}(x|y)=\dfrac{f(x,y)}{f_Y(y)}=\begin{cases}3e^{-3x}, & x>0\\ 0, & x\le 0\end{cases}$

(5) $P\{X>1|Y=2\}=P\{X>1\}=\int_1^{+\infty}3e^{-3x}dx=e^{-3}$. ($\because X$ 与 Y 独立)

(6) $f_Z(z)=\int_{-\infty}^{+\infty}f_X(x)f_Y(z-x)dx=\begin{cases}\int_0^z 3e^{-3x}\cdot 2e^{-2(z-x)}dx=6e^{-2z}(1-e^{-z}), & z>0\\ 0, & z\le 0\end{cases}$

$\begin{cases}x>0\\ z-x>0\end{cases}\Rightarrow$ 积分区域为



4. (1)

X	1	2	3
p_k	0.25	0.3	0.45

Y	-1	0	3	5
p_k	0.35	0.25	0.2	0.2

$E(X)=1\times0.25+2\times0.3+3\times0.45=2.2$. $E(Y)=(-1)\times0.35+0\times0.25+3\times0.2+5\times0.2=1.25$

(2) $E(X^2)=5.5$. $E(Y^2)=7.15$. $D(X)=E(X^2)-[E(X)]^2=0.66$. $D(Y)=5.5875$

(3) $Cov(X,Y)=E(XY)-E(X)E(Y)=2.6-2.2\times1.25=-0.15$

$\rho_{XY}=\dfrac{Cov(X,Y)}{\sqrt{D(X)}\sqrt{D(Y)}}=\dfrac{-0.15}{\sqrt{0.66}\sqrt{5.5875}}$

5. (1) $\mu_1=E(X)=\theta+1$. $\therefore \theta=\mu_1-1$. $\hat{\theta}_{矩}=\bar{X}-1$

设样本值为 $x_1,\cdots,x_n$. 似然函数为

$$L(\theta)=\prod_{i=1}^{n}f(x_i)=\begin{cases}\prod_{i=1}^{n}e^{-(x_i-\theta)}, & \theta\le\min\{x_1,\cdots,x_n\}\text{ 时}\\ 0, & \text{否则}\end{cases}$$

$\therefore$ 当 $\theta\le\min_{1\le i\le n}\{x_i\}$ 时 $L(\theta)=e^{-\sum x_i+n\theta}$ 单增. $\therefore \hat{\theta}_{MLE}=\min\{X_1,\cdots,X_n\}$

(2) $E(\hat{\theta}_{矩})=E(\bar{X}-1)=E(\bar{X})-1=E(X)-\theta+1-1=\theta$ $\therefore \hat{\theta}_{矩}$ 是 θ 的无偏估计

$$F(x)=\begin{cases}1-e^{-(x-\theta)}, & x>\theta\\ 0, & x\le\theta\end{cases}\quad\therefore F_{\hat{\theta}_{MLE}}(x)=1-[1-F(x)]^n=\begin{cases}1-e^{-n(x-\theta)}, & x>\theta\\ 0, & x\le\theta\end{cases}$$

$$f_{\hat{\theta}_{MLE}}(x)=\begin{cases}ne^{-n(x-\theta)}, & x>\theta\\ 0, & x\le\theta\end{cases}\quad\therefore E(\hat{\theta}_{MLE})=\frac{1}{n}+\theta\ne\theta\quad\therefore\hat{\theta}_{MLE}\text{ 不是无偏估计.}$$

6. (1) 枢轴量 $\dfrac{(n-1)S^2}{\sigma^2}\sim\chi^2(n-1)$.

(2) 由 $1-\alpha=0.98$. $\therefore\alpha=0.02$. σ^2 置信区间 $\left(\dfrac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}}(n-1)},\dfrac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}\right)=\left(\dfrac{99\times25}{\chi^2_{0.01}(99)},\dfrac{99\times25}{\chi^2_{0.99}(99)}\right)$

$n=100$, $S^2=25$

(3) 证明: $\bar{X}\sim N\left(0,\dfrac{3^2}{9}\right)=N(0,1)$. $\dfrac{(n-1)S^2}{\sigma^2}=\dfrac{9S^2}{4^2}\sim\chi^2(9)$

且两个变量相互独立, 由 t 分布的定义.

$$\therefore\frac{\bar{X}}{\sqrt{\frac{9S^2}{16}/9}}=\frac{4\bar{X}}{S}\sim t(9).$$